

AP Automation - written assignment

Part A — Research & Baseline (20 min)

1) What is the industry benchmark for cost-per-invoice in manual AP processing? (cite your source).

Industry benchmark (cost per invoice):

According to benchmark data compiled from **APQC**, **Ardent Partners**, and **Deloitte**, the cost of manually processing an invoice ranges from **\$12.88 to \$19.83 per invoice**, while AI-driven automation reduces this to approximately **\$2.36 per invoice**.

Sources:- [https://parseur.com/blog/ai-invoice-processing-benchmarks#:~:text=Speed%20\(according%20to%20SuperAGI\),per%20invoice%2C%20according%20to%20Ascend.](https://parseur.com/blog/ai-invoice-processing-benchmarks#:~:text=Speed%20(according%20to%20SuperAGI),per%20invoice%2C%20according%20to%20Ascend.)

2. Name one real AP automation product or platform (not UiPath or RPATech) and state one specific feature it offers for invoice capture or matching.

AP automation product:

One real AP automation platform is **Parseur** (mentioned in the source). It offers AI-powered OCR that extracts invoice data from various formats and automatically validates it before sending it to ERP systems.

3. What does "3-way match" mean in AP, and what are the three documents involved?

A **3-way match** is a verification process in Accounts Payable used to ensure that a vendor invoice is correct before making payment. It helps confirm that the company only pays for what was actually ordered and received.

It involves matching three documents:

- **Purchase Order (PO):** shows what was ordered and at what price
- **Goods Receipt Note (GRN):** confirms what was actually received
- **Vendor Invoice:** shows what the supplier is billing

In this process, key details like **quantity, price, and item description** are compared across all three documents. If everything matches, the invoice is approved for payment; if there is any mismatch, it is flagged for review.

This ensures accuracy, prevents overpayment, and reduces the risk of errors or fraud.

Part B — Process Map (15 min)

Draw or write out the end-to-end AP invoice workflow for MedCare as it exists today (manual). Number each step. Mark with $\triangle$ any step where errors or delays are most likely.

1) **Invoice Receipt** $\triangle$

Invoices are received through email, fax, or supplier portal, often in different formats. This creates risk of missing or untracked invoices.

2) **Invoice Review**

Initial check is done to verify invoice completeness and basic details like vendor name and amount. Errors can occur due to manual checking.

3) **Data Entry in ERP** $\triangle$

Invoice details are manually entered into the ERP system. This step is highly error-prone due to manual typing.

4) **3-Way Matching** $\triangle$

Invoice is compared with PO and GRN to verify quantity, price, and item details. This is complex and time-consuming when done manually.

5) **Exception Handling** $\triangle$

Any mismatch is investigated by contacting vendors or internal teams. This causes major delays due to back-and-forth communication.

6) **Approval Process** $\triangle$

Invoice is sent for manual approval to authorized personnel. Delays happen if approvers are unavailable.

7) **Payment Scheduling**

Payment is scheduled based on agreed terms and due dates. This step is relatively structured.

8) **Payment Processing**

Payment is executed through banking systems. Usually standardized with fewer errors.

9) **Posting & Archival**

Final transaction is recorded in the ERP system. Mistakes here can impact financial records. Invoice and related documents are stored for future reference and compliance. Manual storage can lead to document loss or retrieval issues.

Part C — Automation Proposal (15 min)

From your process map, pick the TWO steps with the highest automation potential.

For each:

- State the step
- Explain why it is automatable (what makes it rule-based or structured enough for a bot/AI?)
- Name the type of technology you would use (e.g. OCR, RPA bot, AI classification, workflow rule).

1) Data Entry in ERP

- **Why automatable:**
Invoice data such as invoice number, date, vendor name, GST, and amount appears in a structured or semi-structured format across most invoices. Since this task involves repetitive extraction and entry of similar fields, it is rule-based and does not require human judgment in most cases.
- **Technology: OCR (Optical Character Recognition) + AI-based Intelligent Document Processing (IDP) .**
- **Impact:** Automatically extracts and inputs data into ERP, reducing manual effort, minimizing typing errors, improving accuracy up to ~99%, and reducing processing time from minutes to seconds.

2) 3-Way Matching

- **Why automatable:**
3-way matching involves comparing predefined fields like quantity, price, and item details across PO, GRN, and invoice. These comparisons follow clear business rules (e.g., quantity must match, price within tolerance), making the process highly rule-driven and suitable for automation.
- **Technology: RPA bot + rule-based workflow engine**
- **Impact:** Enables automatic matching and instant validation, reduces manual checking effort, flags exceptions quickly, and improves speed and consistency while lowering error rates.

Part D — Client Manager Summary (10 min)

Write exactly 3 sentences addressed to the MedCare CFO summarising: what the current problem is, what you are proposing to automate, and what outcome they should expect.

- 1) Currently, MedCare Hospitals is relying on a manual Accounts Payable process, which is increasing processing costs, slowing down invoice cycles, and creating frequent delays due to manual data entry, 3-way matching, and time-consuming exception handling.

2) We recommend automating invoice data capture and 3-way matching using AI, OCR, and RPA so that routine invoices can be processed automatically, with only exceptions requiring human attention.

3) We recommend automating invoice data capture and 3-way matching using AI, OCR, and RPA so that routine invoices can be processed automatically, with only exceptions requiring human attention.